Executive Summary: Campaign Response Targeting

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The problem

The last campaign went to every customer and 15% responded. At a cost of 3 per contact and 11 revenue per response, a campaign needs a **27.3%** response rate to break even. Contacting everyone therefore **lost 2,983**.

The finding

A propensity model (logistic regression, out-of-fold AUC 0.89) ranks customers by their likelihood to respond. Contacting only the top-ranked customers turns the campaign profitable.

Strategy	Contacted	Responders captured	Profit
Contact everyone (actual)	2,208	331 (100%)	-2,983
Top 10% by model score	221	147 (44%)	+954
Top 20% by model score	**442**	**235 (71%)**	**+1,259**
Top 50% by model score	1,105	313 (95%)	+128

Only the top two deciles respond above break-even (66.5% and 39.8%). The bottom 20% of customers (440) produced **zero** responders.

What drives response

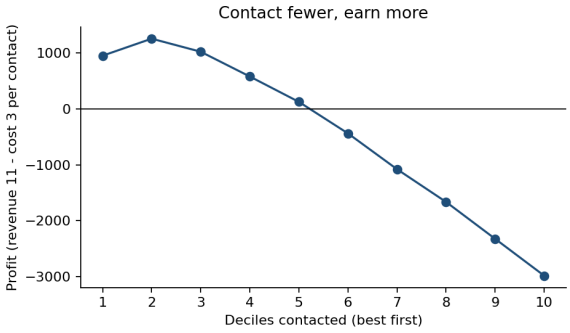
- **Past behaviour is the strongest signal.** Customers who accepted any earlier campaign responded at 41%, against 8% for those who never did (gap 32 points, 95% CI 28 to 37).
- **Recent buyers respond more:** 29% for purchases in the last 20 days, 6% for 81+ days inactive.
- **Households with children respond far less** (about 10%, against 27% with none).
- **Income alone is misleading.** It largely reflects spend; among mid-spend customers, higher income responded *less*. Age, web visits, and complaints showed no significant effect.

Recommendation

For the next campaign, contact the top 20% of customers by model score instead of everyone. Expected result on this data: about 80% fewer contacts, roughly 70% of responders retained, and a swing from loss to profit.

What this analysis cannot tell you

- **Response is not incremental impact.** There was no control group, so many top-ranked customers may have bought anyway. The profit figures measure profit from responders, not profit caused by the campaign.
- **The cost of 3 and revenue of 11 are assumptions** taken from two undocumented columns. A different ratio changes the best contact depth.
- **One campaign, 331 responders.** The 71% capture rate is an estimate and needs confirming on a new campaign.

Next step

Run the next campaign with a randomized holdout: leave a random share of the targeted group uncontacted and include a small random sample of low-scoring customers. This measures true incremental lift and tests the model on new data.